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MILITARY SPECIFICATION

FILTERS, PARTICULATE (HIGH-EFFICIENCY FIRE RESISTANT)

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers eight sizes and eight types of high-efficiency, fire resistane particulate filters for use in air cleaning or air filtering systems (see 6.1).

1.2 Classification. Filters covered by this specification shall be of the following sizes and types (see 6.2).

1.2.1 Sizes.

Size	Nominal rated capacity cubic feet per minute (CFM)
1	25
2	50
3	125
4	500
5	1000
6(M20)	1250
7	1500
8	2000

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPD-TS, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

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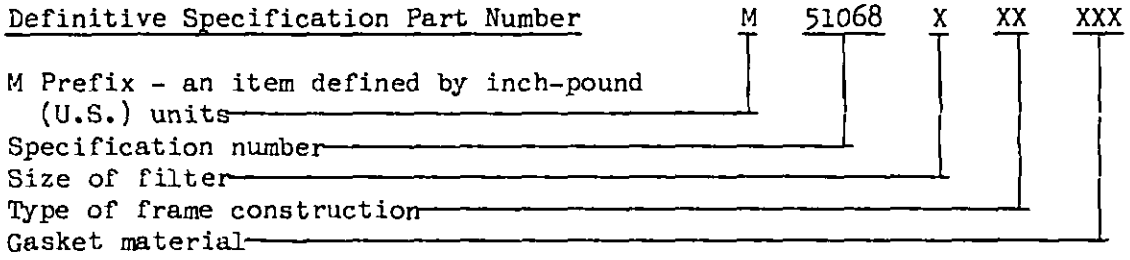
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1.2.2 Types.

Type	Frame Construction (See 3.2.1.1)
IA	Marine Plywood
IB	Exterior Plywood
IC	Wood Particleboard
ID	Laminated Veneer Lumber
IIA	Aluminum Alloy
IIB	Cold Rolled Steel
IIC	Chromized Steel
IID	Stainless Steel

1.2.3 Specification definitive part number. The Specification Part Number is definitive and formatted to identify each item covered by this specification. The Part Number is formatted by selecting from the requirement options available in this specification as follows:



1.2.3.1 Size of filter A one-position alpha field used to designate the size of filter required (table I).

1.2.3.2 Type of frame construction. A 2-position numeric field used to designate the type of frame construction (table II).

1.2.3.3 Gasket material. A 3-position numeric field used to designate the type of gasket material (table III).

TABLE I. Filter size

Part No.	Size	Nominal rated capacity (cfm)
A	1	25
B	2	50
C	3	125
D	4	500
E	5	1000
F	6(M20)	1250
G	7	1500
H	8	2000

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TABLE II. Frame construction

Part No.	Type	Frame construction
10	IA	Marine plywood
20	IB	Exterior plywood
30	IC	Wood particle board
40	ID	Laminated veneer lumber
50	IIA	Aluminum alloy
60	IIB	Cold rolled steel
70	IIC	Chromized steel sheet
80	IID	Stainless steel

TABLE III. Gasket material

Part No.	Material
100	Cork and rubber composition [see 3.2.2(a)]
200	Oil resistant [see 3.2.2(b)]

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-A-250/2 - Aluminum Alloy 3003, Plate and Sheet.
QQ-A-250/8 - Aluminum Alloy 5052, Plate and Sheet.
QQ-A-250/11 - Aluminum Alloy 6061, Plate and Sheet.
QQ-A-1876 - Aluminum Foil.
QQ-P-416 - Plating, Cadmium (Electrodeposited).
QQ-S-766 - Steel Plates, Sheets, and Strip - Corrosion Resisting.
TT-E-527 - Enamel, Alkyd, Lusterless.
TT-P-25 - Primer, Paint, Exterior (Undercoat for Wood, Ready-Mixed, White and Tints).
PPP-B-636 - Boxes, Shipping, Fiberboard.

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MILITARY

MIL-R-6130 - Rubber, Cellular, Chemically Blown.
MIL-C-6183 - Cork and Rubber Composition Sheet; For Aromatic Fuel and Oil Resistant Gaskets.
MIL-S-46055 - Steel, Sheet and Strip, Low Carbon, Chromized.
MIL-F-51079 - Filter Medium, Fire-Resistant, High Efficiency.
MIL-A-81596 - Aluminum Foil for Sandwich Construction.

STANDARDS

FEDERAL

Fed. Std. No. 141 - Paint, Varnish, Lacquer and Related Materials; Methods for Sampling and Testing.
Fed. Std. No. 595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-282 - Filter Units, Protective Clothing, Gas-Mask Components and Related Products: Performance-Test Method.

2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH AND DEVELOPMENT CENTER

76-2-641 - Penetrometer, Filter Testing, DOP, Q107.
136-14-550 - Tester, Filter, Pressure Resistance, Q160.
136-42-850 - Penetrometer, Filter Testing, DOP, Q76.

(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation.

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INTERNATIONAL CONFERENCE OF BUILDING OFFICIALS

Research Committee Report Number 3155, Feb. 1979.

(Request for copies should be addressed to International Conference of Building Officials, 5360 S. Workman Mill Road, Whittier, CA 90601.)

ASTM STANDARDS

A 165 - Electrodeposited Coatings of Cadmium on Steel.
A 176 - Stainless and Heat-Resisting Chromium Steel Plate, Sheet and Strip.
A 240 - Heat-Resisting Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Fusion-Welded Unfired Pressure Vessels.
A 268 - Seamless and Welded Ferritic Stainless Steel Tubing for General Service.
A 366 - Steel, Carbon, Cold-Rolled Sheet, Commercial Quality.
A 740 - Hardware Cloth (Woven or Welded Galvanized Steel Wire Fabric).
B 209 - Aluminum and Aluminum-Alloy, Sheet and Plate.
D 1056 - Flexible Cellular Materials - Sponge or Expanded Rubber.
D 1225 - Test for Flat Crush of Corrugated Paperboard.
D 3359 - Measuring Adhesion by Tape Test.
D 3951 - Standard Practice for Commercial Packaging.
E 84 - Surface Burning Characteristics of Building Materials.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

ANSI STANDARDS

ANSI A208.1 - Mat-Formed Wood Particle Board.
ANSI A199.1 - Construction and Industrial Plywood.
ANSI B132.1 - High Efficiency Air Filter Units.

(Copies of these standards may be obtained from American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Particulate filters furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for the opening of the bids (see 4.2 and 6.4).

3.2 Materials and components.

3.2.1 Frame.

3.2.1.1 Frame material. The frame shall be made from the material as specified for each of the following filter types:

Type IA - 3/4 inch plywood conforming to marine exterior type, grade A-A, of ANSI-A199.1 or grade A-B of the same standard when grade A-A is not available. Either grade shall be fire retardant treated (see 3.2.1.2).

Type IB - 3/4 inch plywood conforming to exterior type, grade A-A, group 1, of ANSI-A199.1 or grade A-B of the same standard when grade A-A is not available. Either grade shall be fire retardant treated (see 3.2.1.2).

Type IC - 3/4 inch wood particle board for exterior application with filler on both faces and conforming to grade 2M-2 of ANSI A208.1 (with a minimum density of 45 pounds per cubic foot).

Type ID - 3/4 inch laminated veneer lumber, for exterior application, with a minimum density of 36 pounds per cubic foot; made of veneers of 1/10 inch or 1/8 inch in thickness and arrayed for lamination with all grains parallel to the length of the member. Laminated veneer lumber shall comply with the requirements of the "Research Committee Report No 3155," Feb 1979 International Conference of Bldg Officials.

Type IIA - Aluminum Alloy 6061-T6, 5052-H32, 5052-H34 or 3003-H14 sheet, 0.080 inch minimum thickness conforming to QQ-A-250/11, QQ-A-250/8, QQ-A-250/2 respectively, or ASTM B209.

Type IIB - Cold-rolled steel sheet, 0.0747 inch (14 gage USS) minimum thickness conforming to ASTM A366. Prior to frame construction, the metal sections shall be cadmium plated in accordance with class 2, type II of QQ-P-416 or ASTM A165, type OS. Welded frame assemblies shall be plated after welding.

Type IIC - Chromized steel sheet, 0.0747 inch (14 gage USS) minimum thickness, conforming to MIL-S-46055.

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TYPE IID - Stainless steel, 0.0747 (14 gage USS) minimum thickness, conforming to QQ-S-766 or ASTM A176 with the exception that the chemistry shall be in accordance with TP-409 of ASTM A268 and TP-304 of ASTM A240.

3.2.1.2 Flame spread classification. The plywood and wood particleboard frames shall have a flame spread classification of 25 or less when tested as specified in ASTM E84.

3.2.2 Gasket material. Unless otherwise specified in the contract or purchase order, (see 6.2), gasket material shall be one of the following materials:

(a) Cork and rubber composition conforming to type I or type II, class 2, grade A, soft of MIL-C-6183.

(b) Oil resistant, expanded cellular rubber in accordance with grade SCE-43 or 44 of ASTM D1056 and also qualified for listing on the Qualified Products List (QPL) for type II, grade A, medium of MIL-R-6130.

3.2.3 Faceguards. Unless otherwise specified in the contract or purchase order, faceguards shall be fabricated from hardware cloth conforming to ASTM A740, 1/4-inch mesh size and 0.0258-inch wire size.

3.2.4 Adhesives and sealants. Adhesives used to splice the medium, fasten gasket to filter face, and sealants or adhesives used to seal the filter pack to the frame shall be self-extinguishing after direct contact with an open flame, as evidenced by testing in accordance with the spot-flame test of ANSI B132.1 (see 6.3).

3.2.5 Filter medium. Unless otherwise specified in the contract or purchase order, the filter medium shall conform to MIL-F-51079.

3.2.6 Separators. Unless otherwise specified in the contract or purchase order the pleats of the filter medium shall be separated by corrugated separators made of one of the following materials (see 6.2):

3.2.6.1 Aluminum alloy. Aluminum alloy 1145-H19 conforming to QQ-A-1876, 3003-H19 or 5052-H39 both conforming to MIL-A-81596, with the exception of the temper in the latter. Each alloy shall have a minimum thickness of 0.0015 inch. Corrugated separators made of aluminum alloy shall have a minimum resistance to compression of 10 pounds per square inch (psi) when tested in accordance with 4.4.4.3.

3.2.6.2 Epoxy-coated aluminum. Aluminum as specified above, coated on both surfaces with a vinyl-epoxy coating 0.8 to 1.2 mils thick. The applied coating shall meet the adhesion requirements, off-gas test, and flexibility test specified in 4.2.3.5. Corrugated separators made of epoxy-coated aluminum alloy shall have a minimum resistance to compression of 10 pounds per square inch when tested in accordance with 4.4.4.3.

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3.2.7 Paint finish for filter frame. Where required (see 3.3.7) the exterior surface of the frame shall receive one coat or primer conforming to TT-P-25, tinted to grey matching color No. 26134, Fed. Std. 595 and two coats of olive drab, No. X-34087 finish conforming to TT-E-527.

3.3 Manufacture and assembly.

3.3.1 Construction. Unless otherwise specified in the contract or purchase order each size of filter shall be constructed to conform to figure 1 (see 6.2). The dimensions for the various filters shall be as specified in table I. The fasteners used in the construction of the filter frames shall be compatible with frame materials (see 3.4.7).

TABLE I. Filter dimensions

Filter size	Dimensions, inches		
	X	Y	Z*
1	8 (+0 - 1/16)	8 (+0 -1/16)	3-1/16 (+1/16 -0)
2	8 (+0 - 1/16)	8 (+0 -1/16)	5-7/8 (+1/16 -0)
3	12 (+0 - 1/16)	12 (+0 -1/16)	5-7/8 (+1/16 -0)
4	24 (+0 - 1/8)	24 (+0 -1/8)	5-7/8 (+1/16 -0)
5	24 (+0 - 1/8)	24 (+0 -1/8)	11-1/2 (+1/16 -0)
6	24 (+0 - 1/8)	24 (+0 -1/8)	11-1/2 (+1/16 -0)
7	24 (+0 - 1/8)	24 (+0 - 1/8)	11-1/2 (+1/16 -0)
8	24 (+0 - 1/8)	24 (+0 - 1/8)	11-1/2 (+1/16 -0)

*Excluding the gasket, no part of the filter shall extend beyond this dimension.

3.3.2 Flatness and squareness. The faces of the frame shall be flat and parallel to within a total allowance of 1/16 inch. The frame shall be square to within a total allowance of 1/8 inch when measured diagonally acrosss the corners of both faces.

3.3.3 Gasket assembly. Unless otherwise specified in the contract or purchase order the gasket shall be fastened to the influent face of the filter frame, and shall be made from the material specified in the contract or purchase order (see 6.2). The gasket shall be 1/4 ± 1/32 inch thick by 3/4 ± 1/32 inch wide and flush with the outer edges of the frame. The gasket shall be either molded in continuous unbroken form, or made from 4 strips joined at the corners by interlocking means, so that no gaps are visible and the joint shall be air tight. The gasket shall be continuously cemented to the face of the filter frame so as to prevent any air leakage between gasket and frame.

3.3.4 Faceguards. Unless otherwise specified in the contract or purchase order faceguards shall be used only on filter sizes 4, 5, and 6 (see 6.2). A faceguard shall be installed on each face of the filter in such a manner as to be essentially flat.

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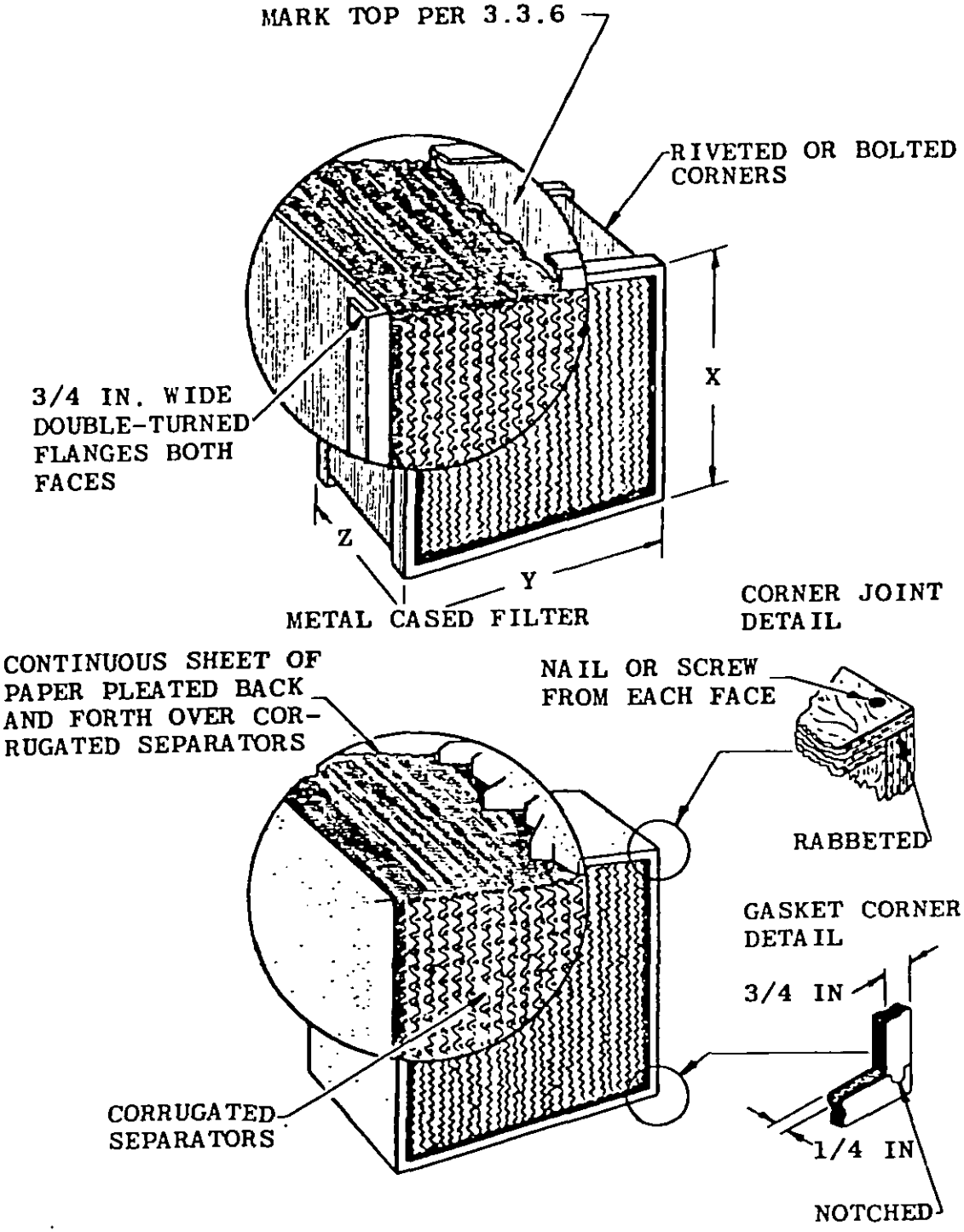


FIGURE 1

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3.3.5 Medium installation. The filter medium shall be securely fastened to sides and ends of filter frame with adhesive to completely seal edges of medium to filter frame. Patching of holes or tears in the medium shall not be permitted.

3.3.5.1 Splices. Splicing of the medium shall not be permitted in sizes 1 and 2. Larger size filters shall not have more than one splice in the filter pack.

3.3.6 Marking. Each filter shall be marked by stencilling or with a non-removable label on the top of the frame to indicate the manufacturer, model, serial number, specification part number, test airflow direction, airflow resistance at rated airflow and the results of the dioctylphthalate (DOP) smoke penetration test at rated airflow and at 20 percent of rated airflow for all sizes of filters. In addition the wording "Install With This Side Up" shall be stencilled on the top of the frame in 3/8 inch minimum height letters. The top of the filter shall be perpendicular to the pleats in the filter medium (see figure 1).

3.3.7 Finish. The exterior surface of type IA, IB, IC or ID (wood) size 6 filter frames shall be painted as specified in 3.2.7. Unless otherwise specified in the contract or purchase order, sizes 1, 2, 3, 4, 5, 7 and 8 shall be unpainted.

3.4 Performance.

3.4.1 DOP smoke penetration. The total DOP smoke penetration through the filter medium, frame, and gasket of a filter which has been encapsulated, shall be no greater than 0.03 percent of upstream concentration at rated aiflow and at 20 percent of rated airflow for sizes 1 through 6 and at 1000 cfm and 20 percent rated airflow for sizes 7 and 8 when tested as specified in 4.4.4.1.

3.4.2 Resistance to airflow. The resistance to airflow at the rated airflow of the filter shall be no greater than 1.0 inch of water for filters, sizes 4 and 5, and 1.30 inches of water for sizes 1, 2, 3, 6, 7, and 8 filters when tested as specified in 4.4.4.1.

3.4.3 Resistance to rough handling. The filter shall be free from cracked or warped frames; loose corners or joints; cracked adhesive; and loose or deformed medium, separators, or faceguards after rough handling as specified in 4.4.4.2 for 15 minutes at 3/4 inch total amplitude and a frequency of 200 cycles per minute with faces, pleats, and separators in a vertical position (see figure 1). Also after rough handling the same filter shall meet the requirements in 3.4.1 and 3.4.2.

3.4.4 Resistance to pressure. Filters, sizes 4, 5, 6, 7 and 8, after being conditioned in accordance with 3.4.4.1 shall withstand the airflow and water spray environment of 3.4.4.2 without rupture of the filter medium; within 15 minutes after completion of the pressure test, and still wet, the filter shall meet the DOP requirement of 3.4.1 at 20 percent airflow.

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3.4.4.1 Conditioning. Prior to being tested for resistance to pressure, the filter shall be conditioned for not less than 24 hours in a chamber at 95° ± 5°F and a relative humidity (RH) of 95 ± 5 percent.

3.4.4.2 Pressure testing. After conditioning as specified in 3.4.4.1, the filter shall be removed from the chamber, wrapped, transported to the test equipment, and tested within 15 minutes in accordance with 4.3.3.1 under the test conditions listed in table II.

TABLE II. Test conditions and requirements

Test conditions	Test requirements
Temperature	95° ± 5°F
Relative humidity	95° ± 5%
*Rate of airborne water droplets flowing toward the filter	1 ± 1/4 pound per minute per 1000 cfm of nominal rated filter capacity
Pressure differential across filter	10.0 ± 0.2 inch of water
Time to reach pressure differential	1/2 minute, maximum
Time duration at sustained water droplet flow and pressure differential across filter	1 hour, minimum
Airflow	That required for producing the above pressure differential

*This is defined as the rate of water flowing through the spray orifice less the fallout and drainage from the air duct walls between points of location of the spray orifice and one inch before the face of the filter.

3.4.5 Resistance to heated air. The DOP smoke penetration through the filter medium, frame and gasket, when tested at the rated airflow of the filter and in accordance with 4.4.4.1 shall not exceed 3.0 percent after heated air at a temperature of 700° ± 50°F has been passed through the filter (face of gasket and sides of filter not exposed) for not less than 5 minutes at the airflow rate for size specified in table III and tested in accordance with 4.3.3.2.

3.4.6 Spot flame resistance. After the removal of the test flame at each point of application, there shall be no sustained flaming on the downstream face of the unit and there shall be no transmittal of flame to the outside surfaces of the frame, gaskets, separators, medium or adhesive when tested in accordance with 4.3.3.3.

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With filter sizes of 1, 2, 3, 4, 7 and 8, a UL label shall be acceptable objective evidence of compliance with this requirement when tested in accordance with the applicable test method in ANSI B132.1.

TABLE III. Airflow rate for heated air

Filter size	Minimum test Airflow at 700° ± 50°F (cfm)
1	25
2	50
3	125
4	500
5	1000
6	1250
7	1500
8	2000

For sizes 1, 2, 3, 4, 5, 7, and 8 (table III) an Underwriter's Laboratories (UL) label shall be acceptable objective evidence for compliance with this requirement when tested in accordance with the applicable test method in ANSI B132.1.

3.4.7 Resistance to environmental exposure. The filters shall show no evidence of corrosion of metal frames, fasteners or faceguard after any cycle and shall comply with the requirements specified in 3.4.7.3 when subjected to the environmental conditions, cyclic exposures, and tested in accordance with 3.4.7.1, 3.4.7.2 and 3.4.7.3.

3.4.7.1 Environmental conditions. The filters shall be subjected, in accordance with 4.3.3.4 to three 3-week cycles of environmental conditions each consisting of one week arctic (-65°F), one week desert (160°F and 10 percent RH) and one week tropical (113°F and 88 percent RH). The time between condition changes shall not exceed 15 minutes. The sequence of environmental changes for each cycle and the number and sequence of cycles shall be as specified in 3.4.7.2.

3.4.7.2 Cyclic exposure. The filters shall be subjected to 3 cycles (a total duration of 9 consecutive weeks) varied as follows:

Cycle 1 - Tropical, arctic, desert
Cycle 2 - Arctic, desert, tropical
Cycle 3 - Desert, tropical, arctic

After each cycle the filters shall meet the requirements specified in 3.4.7.3.

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3.4.7.3 Testing after exposure cycles. The filters shall meet the DOP penetration requirements of 3.4.1 and the airflow resistance of 3.4.2 after cycle 1 and again after cycle 2. After cycle 3 the filters shall meet the rough handling requirements of 3.4.3, the DOP penetration requirements of 3.4.1 and the airflow resistance requirements of 3.4.2.

3.5 Workmanship. The filter shall be free from foreign matter (dirt, oil or viscous material) and damage, such as distorted or cracked frame, deformation or sagging of medium, separators and faceguards, cracks in adhesive and cracks and holes in exposed portions of the medium.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspetion system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Government's responsibility. Unless otherwise specified, the Government will be responsible for the performance of the tests in 4.3. Samples will be forwarded to the laboratory designated by the contracting officer (6.2).

4.1.3 Objective evidence. The contractor shall provide objective evidence including UL labels acceptable to the contracting officer that the requirements of 3.2, 3.4.5, 3.4.6 and section 5 for which specific inspection has not been provided in the specification have been satisfied.

4.1.4 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.3.3 and 4.4.4.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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(a) Qualification inspection (see 4.3)
(b) Quality conformnce inspection (see 4.4).

4.3 Qualification inspection.

4.3.1 Sample. A qualification sample of 11 particulate filters, each manufactured using the same method, material, equipment and processes as will be used during regular production and from the same source of manufacture, shall be submitted to qualify filters' sizes as follows:

(a) Filter size 5 shall qualify filters of sizes 1 through 5 of the same type, material and source of manufacture;
(b) Filter size 6 shall qualify filters of sizes 1 through 6 of the same type, materials and source of manufacture;
(c) Filter size 7 shall qualify filters of size 7 of the same type, material, and source of manufacture; and
(d) Filters of size 8 shall qualify filters of sizes 7 and 8 of the same type, materials and source of manufacture.

When applicable, UL labels shall be affixed to the filters as evidence of conformance with 3.2.1.2, 3.4.5 and 3.4.6 prior to the submission of the filter for qualification.

NOTE: A qualification sample is not required for those filters which were previously qualified and appear in the applicable Qualified Products List, or were tested and approved for inclusion in the list prior to the bid opening date.

4.3.1.1 Samples with splice. If the contractor intends to deliver filters with a splice in the filter medium (3.3.5.1) three of the qualification samples shall contain a splice.

4.3.1.2 Filters with splice. Filters containing a splice shall be clearly marked with the words 'spliced medium' printed with 1/2 inch minimum height letters and located 2 inches below the marking cited in 3.3.6.

4.3.1.3 Material change. Except as noted below, any change in materials or source of manufacture shall require a new qualification sample (4.3.1).

NOTE: When a manufacturer changes his filter medium a new qualification sample shall be required; except: 1) when the replacement medium appears on applicable QPL, or 2) when the replacement medium has been approved for inclusion in the QPL prior to the bid opening date.

4.3.1.4 Reverification of qualification. At least once every 5 years, each manufacturer on the Qualified Products List shall submit samples in accordance with 4.3.1 for requalification.

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4.3.2 Inspection procedures.

4.3.2.1 For examination. Each filter in the qualification sample shall be examined for all the characteristics in the classification of defects (4.4.3.3).

4.3.2.2 For tests. The qualification sample shall be tested for all the requirements of this specification in accordance with the groupings, sizes of filter, quantity and sequence of tests shown in table IV. If applicable (4.3.1.1) one filter in each group A, B and C shall contain a splice and be so indicated.

TABLE IV. Test groups and sequence

Group Quantity Size	Test sequence Requirement	Test paragraph
Group - A Quantity - 4 filters Sizes - 5 or 6	Resistance to rated airflow (3.4.2) DOP smoke penetration at rated airflow or 1000 CFM and at 20% of rated airflow (3.4.1) Resistance to pressure (3.4.4)	4.4.4.1 4.4.4.1 4.3.3.1
Group - B Quantity - 4 filters Sizes - 5 or 6	Resistance to rated airflow (3.4.2) DOP smoke penetration at rated airflow or 1000 CFM and at 20% of rated airflow (3.4.1) Resistance to rough handling (3.4.3) Resistance to heated air (3.4.5)	4.4.4.1 4.4.4.1 4.4.4.2 4.3.3.2
Group - C Quantity - 2 filters Sizes - 5 or 6	Resistance to rated airflow (3.4.2) DOP smoke penetration at rated airflow or 1000 CFM and at 20% of rated airflow (3.4.1) Resistance to environmental exposure (3.4.7.3)	4.4.4.1 4.4.4.1 4.3.3.4
Group - D Quantity - 1 filter Sizes - 5 or 6	Resistance to spot flame (3.4.6) Resistance of separators to compression (3.2.6)	4.3.3.3 4.4.4.3

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4.3.3 Tests.

4.3.3.1 Resistance to pressure. The resistance to pressure for sizes 4, 5, 6, 7 and 8 filters shall be determined with the Q160 Pressure Resistance Tester (Drawing 136-14-550).

4.3.3.2 Resistance to heated air. The resistance to heated air for all sizes of filters shall be determined using a single pass air preheater capable of delivering air at test airflow (table III) and temperature as specified in 3.4.5. In addition, the temperature differential across the face of the filter shall not vary by more than ± 15°F from the mean (see 6.5.2). After the hot air chamber has reached 700° ± 50°F, the filter shall be placed into the chamber, and when the chamber temperature recovers to 700°F ± 50°F, the 5-minute exposure shall commence.

4.3.3.3 Resistance to spot flame. The resistance to spot flame shall be determined in accordance with the applicable test method with ANSI B132.1.

4.3.3.4 Resistance to environmental exposure. The filter shall be exposed in an environmental chamber to the conditions specified in 3.4.7. After cycle 1 and again after cycle 2, the DOP smoke penetration and resistance to airflow shall be determined in accordance with 4.4.4.1. At the end of the third cycle, the filter shall be rough handled and examined for damage in accordance with 4.4.4.2 and the DOP penetration determined in accordance with 4.4.4.1.

4.3.3.5 Coating tests. Coating tests are as follows:

(a) Adhesion test. The coating shall pass an adhesive peel test according to ASTM D3359 using "Method A" with a rating of 3A to be equaled or exceeded.

(b) Off-gas test. Off-gas volatiles evolving from the coating of the separators shall be not greater than 5 percent by weight when a 2 by 2 inch sample of the coated separator is subjected to temperatures from 20°C to 1000°C and the off-gas volatiles are determined by thermo-gravimetric analysis.

(c) Flexibility test. Flexibility test shall be conducted in accordance with Fed. Std. No. 141, Method 6221.

4.3.4 Acceptance/rejection criteria.

4.3.4.1 Qualification examination and tests. Failure of any filter to comply with the requirements of this specification when examined and tested as specified in 4.3.2 shall disqualify the entire sample.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the completed filters of the same size produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions. However, no more than one lot of filter medium shall be represented in any one lot of finished filters.

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4.4.2 Sampling for examination and tests. Sampling for examination and tests and level A packaging shall be conducted in accordance with MIL-STD-105. For major defect 101, resistance to rough handling, level S-4 shall be used.

4.4.3 Inspection procedure.

4.4.3.1 For examination and test. Sample filters shall be examined and tested in accordance with the classification of defects (4.4.3.3) and with MIL-STD-105.

4.4.3.2 For critical defects. Each filter in the lot shall be tested for its critical characteristics listed in the classification of defects.

4.4.3.3 Classification of defects.

(a) Filter, particulate, high-efficiency, fire resistant (figure 1).

Categories	Defects	Acceptance standards
Critical:		
1	DOP penetration (3.4.1)	4.4.4.1
2	Resistance to airflow (3.4.2)	4.4.4.1
Major:	AQL 1.5 percent defective	
101	Resistance to rough handling (3.4.3)	4.4.4.2
	AQL 1.0 percent defective	
102	Component incorrect, missing, or incorrectly assembled (3.3.1)	
103	Dimensions not as specified (3.3.1)	
104	Damage to exposed portions of medium (3.4.3)	
105	Flatness incorrect (3.3.2)	
106	Faceguard incorrect (3.2.3), (3.3.4)	
107	Frame not square within 1/8 inch on diagonal (across face) (3.3.2)	
108	Frame faces not parallel within 1/16 inch (3.3.2)	
109	Filter pack shows patches or more than one splice (3.3.5.1)	

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110	Finish not as specified (3.3.7)
111	Gaskets not as specified (3.3.3)
112	Workmanship (3.5)

Minor: AQL 2.5 percent defective

201	Faceguard not essentially flat (3.3.4)
202	Marking missing or illegible (3.3.6)
203	UL label missing (3.4.5, 3.4.6)

(b) Packaging (section 5).

Categories	Defects

Critical: None defined

Major: AQL 4.0 percent defective

101	Filter not oriented properly in box
102	Filter face not covered as specified
103	Unit packing assembly incorrect
104	Packing assembly incorrect
105	Shipping container construction, closure, and reinforcing taping not as specified
106	Marking missing, incorrect or illegible

4.4.4 Tests.

4.4.4.1 Resistance to airflow and DOP smoke penetration. The resistance to airflow and DOP smoke penetration for sizes 1, 2 and 3 filters shall be determined with the Q76 DOP Penetrometer (Drawing 136-42-850). The resistance to airflow and DOP smoke penetration for sizes 4, 5, 6, 7 and 8 filters shall be determined with the Q107 DOP Penetrometer (Drawing 76-2-641). The recorded pressure drop shall not include the air resistance created by the test apparatus.

4.4.4.2 Resistance to rough handling. The filters shall be rough handled in accordance with test method T-105.10 of MIL-STD-282 and examined for damage and DOP penetration determined in accordance with 4.3.4.1. Filters shall be tested without packing or packaging. The filter alone shall be held firmly on the movable plate resting on one side with its pleats and separators in a vertical position. The clamps which hold the filter firmly to the plate shall not distort the frame or the body of the filter.

4.4.4.3 Resistance of separators to compression. Three strip separators shall be removed from group D filter (11th filter) after the spot flame test has been completed. The strips shall be selected from the untested and undamaged portion of the filter medium. Three 4 by 4 test specimens shall be cut

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from the selected strips using a suitable cutter (see 6.8) and tested for compression resistance in accordance with ASTM D1225 with the provision that the horizontal movement of the specimen shall be restricted. The compressed load shall be applied at a rate of 25 ± 5 lbs force per second until the specimen collapses completely. The collapsing load shall be noted and compared with the requirements for resistance to compression stated in 3.2.6.

4.4.5 Removal from the Qualified Products List. If at any time the Government elects, as set forth in 4.1.1, to test any filter procured under this specification against any of the requirements of this specification, and the filter fails to meet the requirements as determined by Chemical Systems Laboratory (DRSMC-CLW-E(A)) the manufacturer will be notified and removed from the applicable Qualified Products List.

5. PACKAGING

5.1 Preservation. Preservation shall be as specified herein (see 6.2).

5.1.1 Level A (size 6 filters [M20]). The M20 filters shall be preserved and unit packed as specified on the repair parts packaging data sheet which is identified by the National Stock Number.

5.1.2 Level C (size 1 through 8 filters). Filters of one size (see table I) shall be unit packed in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A or commerical (see 6.2).

5.2.1 Level A (size 6 filter [M20]). Each filter packaged as specified in 5.1.1 shall be packed with the filter pleats and separators in the vertical position within a fiberboard box conforming to style RSC, grade V3c (for a type 1 load) of PPP-B-636. Closure shall be in accordance with requirements for closure of class weather-resistant boxes in the appendix of the box specification. In addition, boxes shall be taped in accordance with the requirements for taping style RSC, class weather-resistant boxes to be shipped as exterior containers; and shall be reinforced with tape, in accordance with the applicable requirements in the appendix of the box specification.

5.2.2 Level C (size 1 through 8 filters). Filters of one size shall be unit packed as specified in 5.1 and packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special markings required by the contract or order, marking shall be as specified herein.

5.3.1 Level A packaging. Level A packaging (5.1.1) shall be marked in accordance with the repair parts packaging data sheet which is identified by the National Stock Number. Level A shipping containers (5.2.1) shall be marked in accordance with MIL-STD-129 and shall include the special marking of 5.3.3.

5.3.2 Level C packaging. Level C packaging (5.1.2) and packing (5.2.2) shall be marked in accordance with MIL-STD-129 and shall include the special marking of 5.3.3.

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5.3.3 Special marking. The level C unit containers (5.1.2) and shipping containers (5.2.2) shall be clearly marked with the following information in accordance with the applicable requirements of MIL-STD-129:

(a) Mark on two adjacent sides the word "UP" with an arrow toward the top of the container to indicate vertical position of filter pleats.

(b) "FRAGILE" and "HANDLE WITH CARE" labels and/or markings shall be applied to two adjacent sides of the box.

(c) Specification part numbers for items described in this specification shall be formatted as shown in 1.2.3 and shall be shown in exterior identification blocks.

5.4 Unitization. When specified in the contract or order, filters, packed as specified shall be palletized in accordance with MIL-STD-147.

5.5 Repair part (size 6 filter [M20]). When this item is procured for storage and issue as a repair part, preservation, unit packing, packing, and marking shall be as specified on the packaging data sheet which is identified by the National Stock Number.

6. NOTES

6.1 Intended use. Each filter covered by this specification is intended to be used as a particulate filter, singly or in parallel in air purifiers or air filtering equipment that will be used in air at temperatures up to 250°F and which may contain entrained moisture.

6.2 Ordering data. Acquisition documents should specify:

(a) The title, number and date of this specification.
(b) Specification part number (see 1.2.3).
(c) Gasket material location, if other than specified in 3.3.3.
(e) Faceguard requirement if other than specified in 3.3.4, and material 3.2.3.
(f) Construction if other than specified in 3.3.1.
(g) Finish if other than specified in 3.3.7.
(h) The level of preservation and packing.
(i) The laboratory to which samples are to be sent for testing.
(j) Filter medium, if other than specified in 3.2.5.
(k) Separator requirement, if other than specified in 3.2.6.
(l) Quantity of packages per shipping container.
(m) Separator material (3.2.6).
(n) Unitization required (5.4).

6.3 Adhesives and sealants. Any adhesive or sealant found suitable during the qualification tests will be acceptable.

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6.4 Qualification. With respect to products requiring qualifications, awards will be made only for such products as have, prior to the bid opening date, been tested and approved for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders on the products covered by this specification. Information pertaining to qualification may be obtained from Commander, US Army Armament Research and Development Command, ATTN: SMCCR-PPP, Aberdeen Proving Ground, MD 21010-5423. Samples shall not be shipped until the above addressee has furnished written shipping instruction to the manufacturer.

6.5 Heated air testing.

6.5.1 Heated air tester. A suitable tester which is currently used at the Edgewood Area of Aberdeen Proving Ground consists of the following components: an air supply fan (1500 CFM @ 700°F, 4" S.P.), a gas fired air heater (700,000 Btu/hr), an air mixer (baffles and louvers), a test chamber, and an exhaust duct. Liquid propane gas (ca. 2500 Btu/ft.^{3}) is used as a fuel for the heater and the exhaust is vented to the outside. Both the airflow and the temperature controls are adjustable. The holding fixture for the test filter shall permit all the air to pass through the filter and shall prevent the air from passing over the face of the filter gasket and sides of the filter. This tester is available for inspection at Product Assurance Directorate, US Army Armament, Munitions and Chemical Command, Edgewood Area, Aberdeen Proving Ground, MD 21010-5423.

6.5.2 Temperature differential determination. The following procedure has been found to be satisfactory. In order to determine temperature differential at the filter's face, divide the filter's face into 36 imaginary equal areas. Determine the temperature at the center of each area. The temperature so determined shall be averaged. The temperature reading of each area shall not differ from the average by more than 15 degrees (°F).

6.6 Compression testing. Suitable test equipment for measuring resistance to compression is: Instron Model 1101 tensile tester equipped with a variable speed range, Instron number 2511-202 compression load cell, 50 pounds maximum, Instron number 2501-025 compression anvil and cable, Instron number A6-17, 6-inch diameter compression plate, coupled to a high speed graphic recorder. Also, a 4-inch square sample has been found to be suitable for performing this test.

6.7 Specimen cutter. A satisfactory cutter for the preparation of the compression test specimens is the Elmendorf Sample Cutter 60-92, manufactured by the Thining Albert Inst. Co., 10960 Dutton Road, Philadelphia, PA.

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6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - EA
Air Force - 99

Review activities:

Army - MD, ME
Navy - AS
Air Force - 84

User activities:

Navy - MC, SH

Civil agencies interest:

DOE-OS (ISFP)

Preparing activity:

Army - EA

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-51068F	FILTERS, PARTICULATE (HIGH-EFFICIENCY FIRE RESISTANT)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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